

Forest Fires Prevention

Final Project Report

1. Introduction

The purpose of this project was to analyze the Forest Fires dataset and identify the environmental conditions associated with wildfire occurrence and severity. The project also evaluated several regression and classification models to determine their suitability for predicting wildfire behavior. The overall objective was to provide evidence-based recommendations that can assist forestry managers in planning resource allocation and wildfire prevention strategies.

2. Dataset Description

The dataset was obtained from the UCI Machine Learning Repository and contains meteorological measurements, Fire Weather Index (FWI) variables, spatial coordinates, and the burned area for each recorded fire. The dataset initially contained 517 observations and 12 predictor variables. No missing values were detected. Four duplicate observations were identified and removed, resulting in a final dataset of 513 observations used throughout the analysis.

3. Exploratory Data Analysis

Exploratory data analysis included examination of the dataset structure, summary statistics, duplicate records, distributions, outliers, correlations and categorical patterns. Most numerical variables exhibited right-skewed distributions, particularly the burned area variable. Correlation analysis showed weak linear relationships with burned area, with temperature displaying the strongest positive correlation and relative humidity showing a weak negative relationship. These findings suggested that wildfire size is influenced by several interacting factors rather than a single dominant predictor.

4. Regression Modelling

Five regression models were investigated: a baseline multiple linear regression, a nonlinear model with polynomial terms, a model including interaction effects, a model with indicator variables for categorical predictors, and a log-transformed model. The baseline model provided the best balance between simplicity and statistical performance. Although statistically significant, the model achieved a relatively low R^2 , indicating that only a small proportion of the variability in burned area could be explained by the available predictors.

5. Model Diagnostics

Model assumptions were evaluated using residual plots, Q-Q plots, Cook's Distance, and the Variance Inflation Factor (VIF). Residual plots indicated substantial variability, while the Q-Q plot suggested departures from perfect normality. Cook's Distance identified eight influential observations deserving further investigation. All predictor VIF values were below 5, demonstrating that multicollinearity was not a major concern.

6. Regularization

Ridge and Lasso regression were applied to evaluate whether regularization could improve model generalization. Both approaches produced nearly identical coefficient estimates and test performance. Lasso achieved a marginally lower Mean Squared Error, but the improvement was negligible. Since VIF values were already low, the limited benefit of regularization was expected.

7. Classification Analysis

The continuous burned area was converted into a binary target representing low-risk and high-risk fires using the median burned area as the threshold. Predictors were standardized before fitting a logistic regression model. The classifier achieved an accuracy of 50.49%, while precision, recall and F1-score were all close to 0.50. These results indicate that the selected predictors alone were insufficient for reliable wildfire risk classification.

8. Overall Comparison

The baseline regression model was preferred because it combined simplicity, interpretability and the strongest statistical performance among the regression models. More complex models increased complexity without meaningful gains. Similarly, logistic regression offered a straightforward classification framework but its predictive accuracy remained limited.

9. Conclusions and Recommendations

The analysis demonstrates that temperature, DMC, DC, relative humidity and location contribute to wildfire behaviour, although their explanatory power is limited. Forestry agencies should monitor these indicators as part of early warning systems while recognizing that additional information such as vegetation type, fuel load, elevation, topography and human activity is likely required for highly accurate prediction. Future work should explore richer datasets and more advanced machine learning techniques to improve predictive performance.